Experiment - 05

Problem:

We are given initial memory with n string and we are given queries 'm'. We have to check that for each query :-
Do we have any string in the memory which is of the same length as queries [i] and differs only in 1 character.

Algorithm.

query → strings.

1. for every index i in s.
   - ↳ Store the original character
   - ↳ Try replacing it with 'a', 'b' and 'c' (except the original character.)
2. After Replacing :
   - ↳ check if the new string exists in hash set.
   - ↳ If yes → print "YES" and stop checking.
3. Restore the original character.
4. If no match found after all positions
   - ↳ print "NO".

Complexity

Building set - $O(n)$

Each Query - $O(L)$

Total $\Rightarrow O(n+L)$

Worst case → $O(nL)$.

```
#include <bits/stdc++.h>
using namespace std;
int main()
{   int n, m;
    cin >> n >> m;
    ~~unordered_map~~
    unordered_set <string> st(n);
    for (int i=0; i<n; i++)
    {
        string s;
        cin >> s;
        st.insert(s);
    }
    vector<string> queries(m);
    for (int i=0; i<m; i++)
    {
        string s;
        cin >> s;
        bool found=false;
        for(int j=0; j<s.size(); j++)
        {   char original = s[i];
            for (char c: {'a', 'b', 'c'})
            {   if c==original) continue;
                s[i] = c;
                if (st.count(s)) {
                    found = true;
                    break;
                }
            }
            s[i] = original;
        }
        cout << (found? "YES\n": "NO\n");
    }
}
```

abc → 123

1+2+3

Eg - Query - "abc".

$i=0 \Rightarrow$ replace a with b $\Rightarrow$ "bbc".

for c $\Rightarrow$ "cbc".

$i=1 \Rightarrow$ "aac"

$\Rightarrow$ "acc"

$i=2 \Rightarrow$ replace c with a $\Rightarrow$ "aba"

$\Rightarrow$ "abb"

Now, at each step check in the set which contains the original initial memory.

Eg - 2 3

aaaaa

acacaca

Set (store).

acacaca

aaaaa

Query -

① aabaa $\rightarrow$ ✗

$i=0 \Rightarrow$ babaa $\rightarrow$ ✗

$i=1 \Rightarrow$ abbaa $\rightarrow$ ✗

$i=3 \Rightarrow$ aaaaa $\rightarrow$ ✓ (accepted) YES

② caaac $\rightarrow$ ✗

$i=0 \Rightarrow$ aaaac $\rightarrow$ ✗

$\Rightarrow$ baaac $\rightarrow$ ✗

$i=1 \Rightarrow$ ccaac $\rightarrow$ ✗

cbaac $\rightarrow$ ✗

⋮

Check for all the index. then ultimately no string is found which satisfies the condition.